

# What is an API?

An **API (Application Programming Interface)** is a set of rules that allows different software applications to communicate with each other. It works as a **bridge** between the client (like a mobile app or website) and the server (database or backend).

Simply: An API is like a **waiter in a restaurant**.

- The **user** (customer) makes a request.
- The **API** (waiter) delivers the request to the **server** (kitchen).
- The server prepares the response (food) and the API delivers it back to the user.



# Types of APIs

1

## Web APIs

Work over the internet using HTTP/HTTPS (most common, REST, GraphQL).

2

## Library APIs

Provided inside programming libraries (e.g., Dart math library).

3

## Operating System APIs

Allow apps to interact with the system (Android or iOS APIs).

# REST API (Most Common)

REST APIs use standard **HTTP methods** to perform operations:

- GET → Retrieve data.
- POST → Send new data (e.g., user registration).
- PUT / PATCH → Update existing data.
- DELETE → Remove data.

## Example (Login API)

👉 **Request:**

```
POST /login { "username": "rana", "password": "123456" }
```

👉 **Response:**

```
{ "status": "success", "token": "abc123xyz" }
```

# Advantages of Using APIs



## Reusability

Multiple apps can use the same backend.



## Separation of Concerns

Frontend and Backend are independent.



## Security

Access controlled with Authentication & Tokens.



## Scalability

Easy to expand and support multiple platforms (mobile, web, desktop).